→ Maximum Subarray Sum

arr =	2	3	-8	7	-1	2	3	4
	0	1	2	. . . . .				n-1

find starting index (i) to ending index (j)

both i & $j < n$ such that

$$\boxed{\sum_{k=i}^{j} arr[k] \text{ is maximum.}}$$

```
Int function MSS ( ) {
    Currentbest = arr[0] , si = 0 , ej = 0;
    Overall best = arr[0]

    for ( i = 1 ; i < n ; i++ ) {
        Current best = Max ( currentbest + arr[i] , arr[i] ).
                                  Record as ending j      record as starting i
        overall best = Max ( currentbest , overallbest ) ;
    }
    return overall best;
}
```

0	1	2	3	4	5	6	7
2	3	-8	7	-1	2	3	4

CB = 2 OB = 2	i=1	i=2	i=3	i=4	i=5	i=6	i=7
si = 0 ej = 0	CB = 5 ej = 1 OB = 5	CB = -3 ej = 2 OB = 5	CB = 7 ej = 3 si = 3 OB = 7	CB = 6 ej = 4 OB = 7	CB = 8 ej = 5 OB = 8	CB = 11 ej = 6 OB = 11	CB = 15 ej = 7 OB = 15

$$\boxed{\begin{array}{l}\text{Max Sum} = 15 \\ \text{from index} = 3 \text{ to } 7\end{array}}$$

CamScanner